Warm-Up Answer Key

Statistics and Data Displays

LESSON 2-1 · UNDERSTAND STATISTICAL QUESTIONS

MOVE *Order a short list least to greatest / Say what one tick mark on a number line is worth*
SLIDE 1) 3, 5, 8, 11 2) 5 3) 23 4) 87
MODEL 9 12 2

1. 2, 4, 6, 8, 9
2. 4, 8, 11, 15, 21
3. 7, 12, 18, 21, 25, 30
4. 5
5. 10
6. 3
7. 30
8. 0, 0, 1, 1, 1, 1, 2, 2, 3; 1 shows up most

DROPPED *skill 3 – tallying is incidental here; the data sets on the slide and worksheet are already written as numbers, so counting marks never gates the work*

LESSON 2-2 · REPRESENT AND DESCRIBE DATA IN A HISTOGRAM

MOVE *Sort values into given intervals and count each bin / Read a scale that counts by 2, 5, or 10*
SLIDE 1) 10-19 2) 2 3) 30 4) 20
MODEL 2 3 6

1. 2
2. 2
3. 40-49
4. 2
5. 4, 3, 1
6. 12
7. 4
8. 1, 3, 4, 5

LESSON 2-3 · DESCRIBE THE DATA USING THE MEDIAN

MOVE *Order a list, then count in from both ends to the middle value / Find the number halfway between two numbers*
SLIDE 1) 4, 6, 9, 11, 15 2) 4th 3) 14.5 4) 19
MODEL 15 9 14.5

1. 8
2. 11
3. 12
4. 27
5. 18.5
6. 8.5
7. 11
8. 14.5

LESSON 2-4 · REPRESENT AND DESCRIBE DATA IN A BOX PLOT

MOVE *Order a set, find the median, and split it into a lower half and an upper half / Place a number on a scaled number line*
SLIDE 1) 9 2) 2, 4, 6 3) 35 4) 30 and 40
MODEL 11 6.5 16

1. 10
2. 4.5
3. 5, 9, 12
4. 27.5
5. 20 and 25
6. 60 and 70
7. 13
8. median 30, lower half 20, upper half 40

DROPPED *skill 3 – subtracting two-digit numbers is the range and IQR move, and that is Lesson 2-5; building a box plot only asks students to locate five values, not to subtract them*

LESSON 2-5 · DESCRIBE DATA BY RANGE AND INTERQUARTILE RANGE

MOVE *Pick the two numbers a spread needs, then subtract*
SLIDE 1) 19 2) 353 3) 8 4) 20 – 6
MODEL 15 6 11

1. 16	**5.** 12.5
2. 26	**6.** 63
3. 43	**7.** 41
4. 17	**8.** range 22, IQR 13

DROPPED *skill 4 – deciding which spread is larger is what this lesson's second session teaches; once both differences are computed the comparison is not a separate fluency skill*

LESSON 2-6 · MEDIAN AND OUTLIERS / COMPARING DISTRIBUTIONS

MOVE *Find the median, then find it again after the set changes*
SLIDE 1) 7 2) 6.5 3) < 4) 90
MODEL 7 8 6.5

1. 9	**5.** 9
2. 7	**6.** 9
3. 16	**7.** 14
4. 15.5	**8.** 72, then 71.5

DROPPED *skill 3 – comparing with < and > is already inside ordering a list; it adds no separate practice; skill 4 – judging that a value sits far from the rest is this lesson's own content, the definition of an outlier, so it is not a prerequisite the worksheet should assume*

LESSON 2-7 · DIVIDE DECIMALS USING AN ALGORITHM

MOVE *Divide whole numbers / Multiply or divide by 10, 100, or 1000 by moving the decimal point*
SLIDE 1) 8 2) 21 3) 60 4) 2.4
MODEL 60 2.4 0.6

1. 12	**5.** 32
2. 1.2	**6.** 0.256
3. 35	**7.** 7
4. 45	**8.** 1.55

DROPPED *skill 4 – lining up places and annexing zeros to keep dividing is the decimal-division algorithm this lesson teaches, so it is the lesson's content rather than a prerequisite*

LESSON 2-8 · DESCRIBE DATA USING THE MEAN

MOVE *Add the values, count them, and divide / Find the missing addend that reaches a given total*
SLIDE 1) 60 2) 42 3) 6 4) 12
MODEL 60 5 12

1. 10	**5.** 14
2. 13	**6.** 63
3. 22	**7.** 60
4. 40	**8.** mean 17; the seventh score must be 24

LESSON 2-9 · DESCRIBE DATA BY MEAN ABSOLUTE DEVIATION

MOVE *Find how far a value is from the mean, then add the distances*
SLIDE 1) 10 2) 6 3) 12 4) 2.4
MODEL 6 6 12

1. 6

2. 6

3. 0

4. 5

5. 3, 0, 4

6. 20

7. 33 and 47

8. mean 12; distances 4, 2, 0, 2, 4; mean distance 2.4

DROPPED *skill 1 — finding the mean is Lesson 2-8's move, drilled the lesson before, and this lesson's own quick check hands students the mean instead of asking for it; skill 4 — dividing a total by the number of values is that same 2-8 move; doing it to a total of distances is this lesson's definition of the MAD, so it is content, not a prerequisite*

LESSON 2-10 · CHOOSE APPROPRIATE MEASURES

MOVE *Find the mean and the median of the same short list / Spot the value far from the rest, and say which measure it moves*
SLIDE 1) 4 2) 50 3) stretched to the right 4) mean
MODEL 3 12.4 mean

1. median 2, mean 10

2. median 6, mean 17

3. median 14, mean 14

4. 100

5. the median

6. either one; both are 24

7. new mean 12; the median stays 6

8. the median, 207.5; the one price of 1,000 pulls the mean up to 284.5

DROPPED *skill 3 — reading a display as symmetric or skewed was rehearsed in 2-1 and 2-2, and here it is a judgment the lesson builds rather than arithmetic that gates choosing a measure; the worksheet gets at the same idea through the numbers instead*